

Ex: P_2 "is the same as" $\mathbb{R}^3$ - equivalent but not canonically equivalent $\Rightarrow$ depends on choice of basis

Let's simply think of a choice of basis:

Ex: V is a vector space over $\mathbb{R}$ w/ $\dim V = 4$ $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$

Ex: $L: P_2 \rightarrow P_1$ $\underset{\cong \mathbb{R}^3}{P_2} \underset{\cong \mathbb{R}^2}{P_1}$ $L(x^2) = 0$ $L(x) = x$ $L(1) = 1$

$L(3x^2 + 2x + 1) = 0 + 2x + 1$ if $B_1 = [x^2, x, 1]$, basis for P_2

$B_2 = [x, 1]$, basis for P_1

$$L\left(\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad L\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

for x^2

$$L(x) = x \quad L\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad L(1) = 1 \quad L\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$L(ax^2 + bx + c) = bx + c \quad L\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} b \\ c \end{bmatrix}$$

$$L\left(a\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = b\begin{bmatrix} 1 \\ 0 \end{bmatrix} + c\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\underbrace{\begin{matrix} \text{basis vector 1} & \text{basis vector 2} & \text{basis vector 3} \end{matrix}}_{\text{relative to } B_1} \quad \underbrace{\begin{matrix} \text{basis vector 1} & \text{basis vector 2} \end{matrix}}_{\text{relative to } B_2}$

Notation: when we know $L: A \rightarrow B$ on a basis for A and we express the output in

a basis for B , we write L in the following way

$$\begin{bmatrix} | & | & | \end{bmatrix} \begin{bmatrix} \end{bmatrix}$$

Where the first basis vector goes 2nd basis vector input to L

Ex: when $L: P_2 \rightarrow P_1$

$$\begin{matrix} \swarrow & \searrow \\ [x^2, x, 1] & [x, 1] \end{matrix}$$

$$L(x^2) = 0 \quad L(x) = x \quad L(1) = 1$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$